

W019 Strain Build List

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Forty-five strains to construct: 25 double knockouts and 20 triple knockouts. Every one of them is made from a strain that already exists, so no new single knockout is needed and no cross waits on another. Tables 3 and 4 are the build list. Tables 1 and 2 are the strains in hand, and Table 5 is the plate they are all measured on.

Tables generated by `experiments/W019-echo-crispr-array/scripts/build_list_tables.py`

Working notes: `notes/experiments.W019-echo-crispr-array/*.md`

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1 Terms ✗

boot SE The standard error of a strain's mean fitness, from resampling the replicate plates that strain was measured on. It is smaller than *plate SD* because it is the error on a mean rather than the spread of the values that mean was taken over.

fitness Colony growth relative to the wild type on the same plate, so 1 is wild-type growth.

plate SD The standard deviation of a strain's fitness across the replicate plates it was measured on.

routes How many different ways the bench can make a triple from strains that already exist. Every triple is built by crossing one existing double with the remaining single. *routes* 1 means only one of the triple's three pairs is in the freezer, so there is a single way in, and a failed cross blocks that triple until the matching new double from Table 3 is done. *routes* 2 means two of its pairs exist, so a failed cross has an immediate second path.

tier How a double was classed when run 4 was designed. **validation** reproduces a measurement someone has already published, **coverage** was added so that later triples would have a parent to be built from, and **novel** had not been measured by anyone.

2 Strains that already exist ✗

Twelve singles and thirteen doubles were built and measured in run 4. No triple has been constructed in any round.

Table 1: Single knockouts that already exist, all twelve measured in run 4. The three columns under *Ours, run 4* are this lab’s measurement; the two under *Costanzo 2016* are the published single-mutant fitness and its standard error.

id	ORF	common	Ours, run 4			Costanzo 2016	
			fitness	boot SE	plate SD	SMF	SE
s1	YJR060W	CBF1	0.5348	0.0506	0.1076	0.5900	0.1138
s2	YPL081W	RPS9A	0.9007	0.0515	0.1105	0.9550	0.1553
s3	YDR057W	YOS9	1.2430	0.1282	0.2690	1.0435	0.0535
s4	YPL046C	ELC1	1.2066	0.0495	0.1052	1.0433	0.0643
s5	YBR203W	COS111	1.1292	0.0679	0.1426	1.0370	0.0458
s6	YGL087C	MMS2	1.1628	0.0329	0.0706	0.9960	0.0940
s7	YLR104W	LCL2	1.1748	0.0564	0.1187	1.0322	0.0453
s8	YLL012W	YEH1	1.2604	0.0659	0.1402	0.9925	0.0804
s9	YER079W	–	1.3333	0.1504	0.3162	1.0387	0.1173
s10	YKL033W-A	–	0.9251	0.0445	0.0954	1.0327	0.1008
s11	YLR313C	SPH1	1.0882	0.1471	0.3083	0.9843	0.0271
s12	YLR312C-B	–	0.8390	0.0661	0.1397	1.0845	0.0441

SPH1 is an open question. s11, SPH1 (*YLR313C*), is a built single that is part of no triple, at a measured fitness of 1.0882 (Table 1). It can ride along on the measurement plate as a near-wild-type control, or it can be dropped. If it rides along, the plate carries 71 strains rather than 70, at the same five wells each.

Table 2: Double knockouts that already exist, all thirteen measured in run 4. Column attribution is as in Table 1, with the published double-mutant fitness and its standard deviation under *Costanzo 2016*. *parent* says whether the pair is the starting strain for one of the triples in Table 4. A dash in the Costanzo columns means no one has published the pair.

id	pair	Ours, run 4			Costanzo 2016			
		fitness	boot SE	plate SD	DMF	SD	tier	parent
d1	YDR057W + YPL081W	0.6071	0.0479	0.1029	1.0351	0.0447	coverage	yes
d2	YPL046C + YPL081W	0.9335	0.0040	0.0084	–	–	novel	no
d3	YER079W + YPL081W	0.9496	0.0556	0.1195	0.8625	0.0975	validation	no
d4	YLR312C-B + YPL081W	0.6480	0.0454	0.0968	1.0165	0.0418	coverage	yes
d5	YDR057W + YGL087C	0.7884	0.0007	0.0015	1.1372	0.0238	validation	yes
d6	YDR057W + YER079W	0.9403	0.0472	0.0991	1.0963	0.0541	coverage	no
d7	YDR057W + YLR312C-B	0.8895	0.0543	0.1150	1.1783	0.0257	validation	no
d8	YJR060W + YPL046C	0.5649	0.0435	0.0930	0.9663	0.0178	coverage	yes
d9	YBR203W + YPL046C	0.8882	0.0232	0.0501	1.1179	0.0171	coverage	yes
d10	YDR057W + YLL012W	0.9315	0.0269	0.0572	0.9995	0.0690	coverage	yes
d11	YLL012W + YPL046C	0.9087	0.0348	0.0744	1.0251	0.0534	coverage	yes
d12	YER079W + YJR060W	0.6886	0.0650	0.1365	0.6100	0.0183	validation	no
d13	YER079W + YLR312C-B	0.7728	0.0514	0.1084	1.0994	0.0393	coverage	yes

All thirteen doubles go on the measurement plate. The pairs marked **yes** under *parent* are the starting strains for Table 4; the rest are re-measured alongside them, at no cost in construction work and for a second reading on each of those pairs.

One designed double was never built. *YJR060W* $\times$ *YKL033W-A* does not exist: the transformation produced no colonies. Both parent singles exist and were measured. The pair is not part of the build list, and nothing in Tables 3 or 4 depends on it.

3 The 25 doubles to construct $\times$

Cross the two singles named in the row. Both parents are already in the freezer (Table 1) and no row depends on another, so all 25 can be started at once. Each row gives the ORF and, where the gene has one, the common name in parentheses.

Table 3: The 25 double knockouts to construct, one per row. Each row names the two singles to cross, both of them in Table 1.

ID	gene 1	gene 2
D01	YBR203W (COS111)	YDR057W (YOS9)
D02	YBR203W (COS111)	YGL087C (MMS2)
D03	YBR203W (COS111)	YJR060W (CBF1)
D04	YBR203W (COS111)	YKL033W-A
D05	YBR203W (COS111)	YLL012W (YEH1)
D06	YBR203W (COS111)	YLR104W (LCL2)
D07	YBR203W (COS111)	YLR312C-B
D08	YBR203W (COS111)	YPL081W (RPS9A)
D09	YDR057W (YOS9)	YKL033W-A
D10	YER079W	YLL012W (YEH1)
D11	YER079W	YLR104W (LCL2)
D12	YGL087C (MMS2)	YLL012W (YEH1)
D13	YGL087C (MMS2)	YLR312C-B
D14	YGL087C (MMS2)	YPL046C (ELC1)
D15	YGL087C (MMS2)	YPL081W (RPS9A)
D16	YJR060W (CBF1)	YLL012W (YEH1)
D17	YJR060W (CBF1)	YLR104W (LCL2)
D18	YKL033W-A	YLL012W (YEH1)
D19	YKL033W-A	YLR312C-B
D20	YKL033W-A	YPL046C (ELC1)
D21	YKL033W-A	YPL081W (RPS9A)
D22	YLL012W (YEH1)	YLR312C-B
D23	YLR104W (LCL2)	YLR312C-B
D24	YLR104W (LCL2)	YPL046C (ELC1)
D25	YLR104W (LCL2)	YPL081W (RPS9A)

4 The 20 triples to construct $\times$

Cross the existing double in *build from* with the single in *third single*. Start with T01 through T15, which have one route each: a failed cross there blocks the triple until the matching double from Table 3 is finished, so those failures are worth surfacing early. T16 through T20 each have a second route already in hand and can absorb a failed cross without waiting on anything.

Table 4: The 20 triple knockouts to construct, in build order. *build from* names an existing double by ORF and *third single* the remaining gene; crossing the two gives the triple. *routes* counts how many different ways the bench can make the triple from strains that already exist, and is defined in Sec. 1.

ID	gene 1	gene 2	gene 3	build from	$\times$ third single	routes
T01	YKL033W-A	YLR312C-B	YPL081W (RPS9A)	YLR312C-B + YPL081W	YKL033W-A	1
T02	YER079W	YLL012W (YEH1)	YLR312C-B	YER079W + YLR312C-B	YLL012W (YEH1)	1
T03	YLR104W (LCL2)	YLR312C-B	YPL081W (RPS9A)	YLR312C-B + YPL081W	YLR104W (LCL2)	1
T04	YBR203W (COS111)	YLR312C-B	YPL081W (RPS9A)	YLR312C-B + YPL081W	YBR203W (COS111)	1
T05	YER079W	YLR104W (LCL2)	YLR312C-B	YER079W + YLR312C-B	YLR104W (LCL2)	1
T06	YGL087C (MMS2)	YLR312C-B	YPL081W (RPS9A)	YLR312C-B + YPL081W	YGL087C (MMS2)	1
T07	YBR203W (COS111)	YDR057W (YOS9)	YPL081W (RPS9A)	YDR057W + YPL081W	YBR203W (COS111)	1
T08	YDR057W (YOS9)	YKL033W-A	YPL081W (RPS9A)	YDR057W + YPL081W	YKL033W-A	1
T09	YDR057W (YOS9)	YKL033W-A	YLL012W (YEH1)	YDR057W + YLL012W	YKL033W-A	1
T10	YGL087C (MMS2)	YLL012W (YEH1)	YPL046C (ELC1)	YLL012W + YPL046C	YGL087C (MMS2)	1
T11	YBR203W (COS111)	YLR104W (LCL2)	YPL046C (ELC1)	YBR203W + YPL046C	YLR104W (LCL2)	1
T12	YBR203W (COS111)	YKL033W-A	YPL046C (ELC1)	YBR203W + YPL046C	YKL033W-A	1
T13	YJR060W (CBF1)	YLR104W (LCL2)	YPL046C (ELC1)	YJR060W + YPL046C	YLR104W (LCL2)	1
T14	YBR203W (COS111)	YGL087C (MMS2)	YPL046C (ELC1)	YBR203W + YPL046C	YGL087C (MMS2)	1
T15	YBR203W (COS111)	YDR057W (YOS9)	YLL012W (YEH1)	YDR057W + YLL012W	YBR203W (COS111)	1
T16	YDR057W (YOS9)	YGL087C (MMS2)	YPL081W (RPS9A)	YDR057W + YGL087C	YPL081W (RPS9A)	2
T17	YJR060W (CBF1)	YLL012W (YEH1)	YPL046C (ELC1)	YJR060W + YPL046C	YLL012W (YEH1)	2
T18	YBR203W (COS111)	YLL012W (YEH1)	YPL046C (ELC1)	YBR203W + YPL046C	YLL012W (YEH1)	2
T19	YBR203W (COS111)	YJR060W (CBF1)	YPL046C (ELC1)	YBR203W + YPL046C	YJR060W (CBF1)	2
T20	YDR057W (YOS9)	YGL087C (MMS2)	YLL012W (YEH1)	YDR057W + YGL087C	YLL012W (YEH1)	2

5 The measurement plate ✗

Seventy strains on one 384-well layout, five wells each, one pick per strain.

These counts are approximate. They fix how many strains the plate has to carry, not where any of them goes. Three things can still move the arithmetic: the twenty wells held for the wild-type reference are a working figure rather than a fixed allocation, thirteen wells are spare once every strain has its five, and SPH1 may ride along as a seventy-first strain, which the spare absorbs without changing the five. The final well assignment comes from the picklist, not from this table.

Table 5: What goes on the measurement plate, approximately. 70 strains at 5 wells each, one 384-well layout, one pick per strain. Wells per strain is the whole number of times 70 strains divide the 358 wells left after 6 are kept empty to fix the plate’s orientation and 20 are given to the wild-type reference, which gives 5. The 20 wild-type wells are a working figure rather than a fixed allocation, and 13 wells stay spare, so these are strain counts and not a well assignment; the picklist decides that. Every existing double is re-measured alongside the new strains, at no cost in construction work or in wells per strain.

on the plate	strains	wells
singles, already built	11	55
doubles, already built (Table 2)	13	65
doubles, to construct (Table 3)	25	125
triples, to construct (Table 4)	20	100
wild type, BY4741	1	20
total	70	365
orientation blanks	–	6
spare	–	13
plate		384